

The Bare Hand

*Pre-PIE *de- as a free morpheme, the compositional decomposition of *-tero-, and the bimorphemic structure of *deh₃-*

Paper 3 of the der- project. Builds on Chavis & Claude (2026a, 2026b).

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Abstract

The first two papers of this project argued that pre-PIE **der-* “hand, open palm” underlies both the lexical cognate cluster Greek *δῶρον*/Celtic **durno-*/Latvian **dūre* (Paper 1) and the PIE contrastive suffix **-tero-* (Paper 2). The present paper completes the framework by going one step deeper: **der-* itself decomposes into two independently attested morphemes — a bare hand-element **de-* and a locative/spatial extension **-r-* — and the same bare **de-* underlies the give-verb **deh₃-* via a verbal extension **-h₃-*.

The bare **de-* is independently attested as the directional particle **de/do* “to, toward” across the IE family: English *to*, Old Church Slavonic *do*, and most directly as the Greek allative clitic *-δε*, productive in Mycenaean Greek by 1400–1200 BCE (*te-qa-de* “to Thebes,” *a-mi-ni-so-de* “to Amnisos”). The **-r-* extension is independently attested as a locative/spatial morpheme in Lipp’s (2009) reconstruction of athematic locative endings **-er* and **-én* in PIE heteroclitic neuters. The contrastive **-tero-* of Paper 2 decomposes compositionally as **de-* (directional, hand) + **-r-* (locative) + **-o-* (thematic) = “at-the-side-where-one-points”: the spatial-direction adjective that Table 1 of Paper 2 documented across the family. The same **de-* underlies **deh₃-* = **de-* + **-h₃-*: a verbal derivative of the hand-noun whose laryngeal is a derivational extension suffix, not a root-internal constituent of an unanalyzable primitive. **der-* (nominal) and **deh₃-* (verbal) are siblings.

Each morpheme is independently attested; the compositions are morphologically transparent; the resulting semantics emerge directly from the parts. The framework is offered for examination, not as a refutation of the standard view.

1. Introduction

The three-paper project reported in these papers rests on a single substantive proposal: that pre-PIE had a body-part etymon meaning “hand, open palm,” and that this etymon underlies a network of lexical, morphological, and verbal reflexes in Proto-Indo-European whose unity the standard handbooks have not recognized. Paper 1 (Chavis & Claude 2026a) made the case for the lexical branch: the cognate cluster Greek *δῶρον*/Hesychian *δᾱρίϛ*/Celtic **dorn*/Latvian **dūre* reflects a body-part root **der-* “hand.” Paper 2 (Chavis & Claude 2026b) extended the case into morphology: the PIE contrastive suffix **-tero-* descends from the same etymon via the grammaticalization pathway HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE, and the bilateral character documented in every standard reference — “one of a pair as opposed to the other” (Sihler 1995; Fortson 2010) — is a structural consequence of what **de-r-o-* composes to mean.

Paper 3 completes the framework by going one step deeper. The present paper asks: what is **der-* itself? The answer we propose is that **der-* is bimorphemic: a bare hand-element **de-* plus a locative/spatial extension **-r-*. The bare **de-* has its own independent attestation as the directional particle **de/do* “to, toward” preserved across the IE family. The **-r-* has its own independent attestation as a locative morpheme in Lipp’s (2009) reconstruction of PIE heteroclitic locatives. When the two combine — **de-* (hand, directional) + **-r-* (locative) + **-o-* (thematic) — the result is the spatial-direction adjective **de-r-o-* = **-tero-*. Each morpheme surfaces independently elsewhere in PIE; the composition is transparent; the semantics follow directly.

The same decompositional logic extends to the give-verb. **deh₃-* is not a primitive root but a parallel derivative of the same bare **de-*: **de-* + **-h₃-*, where the laryngeal is a derivational verbal extension

suffix rather than a root-internal constituent of an unanalyzable primitive. **der-* (= **de-* + **-r-*) and **deh₃-* (= **de-* + **-h₃-*) are therefore siblings — parallel derivatives of the same bare hand-element, the nominal and verbal branches of a single pre-PIE word for “hand.”

A note on terminology. Throughout this paper **de-* refers to the bare pre-PIE hand-element; **der-* refers to the body-part noun in its bimorphemic form **de-* + **-r-* (the form Paper 1 reconstructed). The notation **t/d-er-* used in earlier versions of this project is retired here. The root is **der-* with original *d-*; the *t-* of the grammaticalized suffix **-tero-* reflects regular suffix-initial devoicing, as argued in Paper 2 §1.

2. The bare **de-* attested in PIE

2.1 The directional particle **de/do* across the family

PIE **de/do* “to, toward” is a well-attested local/temporal particle whose reflexes include Greek *δέ* (originally directional/sequential, later grammaticalized as the connective “but, and”), English *to* (with German *zu*, Dutch *te*) from Proto-Germanic **tō* “to, toward,” Old Church Slavonic *do* “until, up to, toward” (with cognates across Slavic), and Latvian dialectal *da-* directional prefix. Dunkel’s *Lexikon der indogermanischen Partikeln* (2014) treats **de* and **do* as ablaut variants of a single particle, with directional and temporal-sequential functions across the family.

The standard treatments do not provide an etymology for **de/do*. It is listed as a primitive PIE particle — a building block with no further decomposition. The framework proposes the missing etymology: **de/do* is the bare descendant of pre-PIE **de-* “hand,” grammaticalized along the pathway HAND > REACH-EXTEND > DIRECTIONAL. The extended hand points; the word for “hand” is reanalyzed as a directional marker. The pathway is typologically common: Heine & Kuteva (2002) document HAND > LOCATIVE/GENITIVE/POSSESSIVE and related developments across Niger-Congo, Austronesian, Sino-Tibetan, and Mayan. Paper 2’s Table 3 (§9.3) shows that English itself preserves HAND as a living binary-contrast pivot in the idiom “on the one hand ... on the other hand”; the directional use in *to* is the older grammaticalization track, and the two are not unrelated.

2.2 The Greek allative *-δε* and its Mycenaean antiquity

The strongest evidence for the framework comes from Greek, where the bare **de-* element is preserved both as the connective *δέ* (directional/sequential origins) and as the productive allative clitic *-δε*. The clitic attaches to accusatives of inanimate nouns and marks motion toward: *oíkade* “homeward” (from *oἶkos* “house”), *Athénaze* “to Athens” (with sandhi), *pólemonde* “to war” (*Iliad*), *Olympíonde* “to Olympus.” The function is unambiguously directional/allative: motion toward a destination, with no stative or ablative meaning. Hackstein (2018) treats the allative pair ἄλλυδίζ ἄλλη (*Iliad*) as a frozen archaism preserving inherited allative morphology.

The decisive evidence for the framework’s chronological claim comes from Mycenaean Greek. Linear B tablets from approximately 1400–1200 BCE preserve *-de* as a fully productive directional clitic on toponyms, with dozens of attested forms catalogued by Lejeune (1961): *te-qa-de* /Thēg^wasde/ “to Thebes,” *a-mi-ni-so-de* “to Amnisos,” *da-da-re-jo-de* “to the Daidaleion,” among others (van Hes 2021). In some tablets *-de* is written separately from the noun, suggesting clitic rather than fully bound suffix status.

Three features of the Mycenaean attestation matter for the framework. First, **productivity**: Mycenaean *-de* is not a closed set of frozen relics but a productive clitic applying freely across the corpus. This is the morphological status of an active grammatical marker. Second, **depth of grammaticalization**: productive grammaticalization by 1400 BCE implies that the underlying HAND > DIRECTIONAL development must have been substantially complete well before the proto-Greek period, plausibly extending back into PIE itself. Third, **semantic profile**: Mycenaean *-de* is narrowly directional/allative — motion toward, not stative location, not source. This is exactly the semantic profile expected of a morpheme grammaticalized from “where the hand points/extends”: directional, not locative.

2.3 Competing accounts: the *χααί* question and the Greek-Anatolian parallel

Greek *χααί* “on/to the ground” has figured in recent debates about whether to reconstruct an allative case for PIE. Melchert (2017) reads *χααί* as a fossilized allative **ǵmm-éh₂* and argues from it that the Anatolian allative in *-a* descends from PIE **eh₂*. Beekes (2010) and Frisk (1960) treat the form as a petrified locative, consistent with the comparative evidence: Latin *humī* “on the ground” is a locative, Old Prussian *semmai* and Lithuanian *žemai* are locative-stative, Sanskrit *kṣamā* is instrumental. If the locative reading is right, Melchert’s argument for PIE **eh₂* as the Greek-Anatolian allative source is weakened, and the space for a **de--*derived allative is correspondingly wider. We do not settle the question here; we note that it is genuinely open.

Lazzeroni (2006) argues that the Greek *-δε* construction and the Hittite allative in *-a* are part of a shared circum-Aegean linguistic area, both restricted to inanimate referents. Melchert (2017) cites this approvingly. The framework is compatible with either an areal or an inherited interpretation of the Greek-Anatolian parallel. In either case, the convergence of *t/d*-initial directional morphology across the Aegean area — PIE **de/do*, Greek *-δε*, Hittite *-a*, with Hurrian *-da/-šta* as a non-IE areal parallel — demands explanation. The framework provides one: the *t/d*-initial directional element descends from pre-PIE **de-* “hand.”

3. PIE **-r-* as a locative/spatial element

3.1 Lipp’s reconstruction of athematic locatives in **-er* and **-én*

The second morpheme of the decompositional argument is **-r-*. Standard treatments of PIE **-r--* morphemes do not provide a unified etymological account; the *r* is treated variously as a thematic suffix, an extension, or one pole of the heteroclitic **r/n* alternation. Reiner Lipp (2009) provides the key independent attestation.

In the course of reconstructing the PIE heteroclitic **r/n*-stem paradigm, Lipp argues that the locative singular forms of acrostatic heteroclite neuters are built from root nouns combined with an adverbial particle, yielding parallel locative endings **-er* and **-én*. His worked examples include **ud-én/*ud-ér* “in/at the wetness” (from the root **úed-* “wetness, water”), from which the heteroclite neuter **úód-ŕ/*úéd-ŕs* “water” is built; and **Híekw-én/*Híekw-ér* “in/at the lobe” (from **Híekw-* “lobe, swelling”), giving the heteroclite **Híékw-ŕ/*Híékw-ŕs* “liver” (literally: the lobe-location in the body).

On Lipp’s analysis, **-r* and **-én* are not body-part formatives that happen to alternate paradigmatically; they are locative/spatial morphemes meaning “at/in the place of (X).” Body-part terms acquire the **r/n* alternation because body parts are concrete locations on the body. The spatial semantics is primary; the body-part association is downstream. This finding is crucial for the framework: **-r-* in PIE morphology has an independently attested locative/spatial function, and the framework can treat the **-r-* of **-tero-* and **-der-* as the same morpheme Lipp documents in the heteroclitic system — a locative marker meaning “at the place of (X).”

3.2 A complication: not every **-r-* is locative

PIE has **-r--* morphemes that do not appear to have locative semantics: the property/color suffix **-ro-* in **rudh-ró-* “red” and the instrument suffix **-tro-/trom* in **méd-trom* “measure” are relational but not obviously locative. The framework does not require that all PIE **-r--* morphemes be locative; it requires only that the **-r-* of **-der-* and **-tero-* be locative, which Lipp’s reconstruction supports. Whether the property and instrument **-r--* morphemes are homophonous separate morphemes or functional extensions of the same locative **-r-* through different pathways is a question we leave open.

4. The compositional decomposition of **-tero-*

4.1 The composition is morphologically transparent

With both pieces in place — bare **de-* (directional, hand) and **-r-* (locative) — the compositional decomposition of **-tero-* follows directly. The suffix decomposes as **de-* + **-r-* + **-o-* (thematic vowel). Each morpheme is independently attested with the right phonology and the right semantics. The combination follows standard PIE morphological principles: a directional element extended by a locative morpheme and thematized to form an adjective.

The semantics compose transparently. **de-* is “hand,” grammaticalized to “where the hand points.” **-r-* is “at the place of.” The combined **der-* is “at the place where the hand points/extends” — the side toward which one is oriented. The thematized adjective **de-r-o-* is “of/at the side toward which one is oriented”: the spatial-direction adjective documented across the IE family in Paper 2’s Table 1.

A concrete example clarifies the parse. The compound **deks-i-tero-* “right (hand)” (Greek *deksiterós*, Latin *dexter*, Old English *swiþro*) decomposes as: **deks-* (the property word “right”) + *-i-* (linking vowel) + **de-r-o-* (“at-the-side-where-one-points”). The first member specifies which side (the right one); the suffix names the category that side belongs to (the spatial-direction category). Festus’s gloss of *ambidexter* as “having the right (hand) on both sides” is a fossil of this analysis, preserved in Latin literary culture.

4.2 The binary contrast: a structural consequence of the composition

Paper 2 §3 established that the bilateral character of **-tero-* — its function as “one of a pair as opposed to the other” (Sihler 1995; Fortson 2010) — is the mainstream consensus, and that the framework explains where that bilateral character came from. The present compositional analysis completes that explanation at the morphological level.

Bare **de-* is an absolute directional: “toward there,” pointing at a target with no implied alternative and no bilateral frame. Greek *oikade* “homeward” does not presuppose a contrasted alternative direction. The locative **-r-* changes this. It anchors the directional to the body as frame of reference — “at the place of the hand-direction” — and bodies have inherent bilateral symmetry. The question “which side of the body?” applied to an organism with two hands has exactly two answers. The absolute directional becomes a relational one.

The bilateral character of **-tero-* is therefore a structural consequence of what **de-r-o-* composes to mean, not a grammaticalized accident of the suffix’s history. **de-* contributes the directional hand-content; **-r-* grounds it in the body as frame of reference; the body’s bilateral symmetry does the rest. The presence vs. absence of **-r-* is exactly what distinguishes the two paths: **de-r-o-* (with **-r-*) inherits the bilateral frame from the body-locative context; bare **de-* (without **-r-*) skips this filter and stays general-directional. The *ambi-* of *ambidexter* negates a bilateral norm; that norm is what **de-r-o-* built in from the start.

4.3 What the decomposition does for the framework

The compositional decomposition is the strongest single piece of evidence for the three-paper framework as a whole. Paper 1 argued that a body-part root underlies the lexical cognate cluster; Paper 2 argued that the same body-part root underlies the contrastive suffix. The present paper shows that these two claims are not merely compatible but morphologically continuous: the suffix **-tero-* is the hand-noun **de-* with the locative **-r-* and thematic **-o-*, exposed through compositional analysis.

This is a stronger claim than either predecessor made alone. Paper 1’s lexical claim is consistent with the morphological evidence being independent or coincidental. Paper 2’s morphological claim is consistent with the source-noun being unrecoverable. The compositional analysis says: the morpheme **-tero-* is morphologically transparent on the framework’s reading and opaque on the standard reading. Where the standard treatment leaves **-tero-* as a primitive suffix of unknown etymological origin, the framework reads it as a compositional formation whose pieces all surface independently elsewhere in PIE with the right phonology and the right semantics to yield the

composed form.

If the framework is wrong, then **de-* (the directional particle), **-r-* (the locative element), and **-tero-* (the spatial-direction adjective) all happen to share the phonology and semantics for the compositional reading by coincidence. We grant that this is possible. We note only that the coincidence required is substantial: three independent matches of phonology and semantics all converging on the predicted composition. Three independent matches in PIE morphology is a coincidence that asks for explanation. The framework provides one; the standard view does not.

5. **deh₃-* “give” as **de-* + **-h₃-*

The decompositional logic of §4 extends naturally to the give-verb. PIE **deh₃-* “give” is reflected in Sanskrit *dádāti*, Greek *δίδωμι*, Latin *dō/dare*, Old Church Slavonic *dati*, Lithuanian *dúoti*. The reduplicated present *didó-h₃-mi/di-dō-mi* shows the **-h₃-* laryngeal extension clearly: the laryngeal colors the e-grade vowel to o (giving *didō-* < *didóh₃*) and produces the characteristic length alternation of the paradigm.

The framework reads **deh₃-* as **de-* (the bare hand-element of §2) + **-h₃-* (the laryngeal extension), where **-h₃-* converts the bare hand-noun into a verbal stem. The semantics compose transparently: hand + (verbal extension) = the act of handing over = to give. The pathway from a hand-noun to a give-verb is one of the most natural grammaticalizations possible — to hand something is to give it — and Heine & Kuteva (2002) document HAND > GIVE-VERB in numerous languages. What the framework adds to the standard treatment is the etymological motivation for the **de-* element. The standard handbooks treat **deh₃-* as a primary verbal root with **de-* as a root morpheme of unknown deeper structure. The framework treats **de-* as the inherited hand-noun, with **deh₃-* being its verbal derivative.

The nominal/verbal distinction is morphologically consequential and directly resolves what earlier versions of Paper 1 treated as a cost. **der-* (= **de-* + **-r-*) is a noun; *δίδωμι* and its cognates are built on **deh₃-* (= **de-* + **-h₃-*), the verbal derivative. The /i/-reduplication of *δίδωμι* reflects the laryngeal-final verbal stem *deh₃-* exactly as the standard account predicts. Paper 1 treated the /i/-reduplication as a cost of reconstructing **der-* as the verbal base; on the present architecture that treatment was mistaken. **der-* was never the verbal base. The /i/-reduplication belongs to **deh₃-*’s morphology and is not a problem for the framework.

The framework’s claim about **deh₃-* is morphological, not phonological. We do not propose that **h₃* is absent from the verbal stem — the Anatolian evidence for its effects (Kloekhorst 2013; Norbruis 2021) is real, and we accept it. We propose that **h₃* was not always there, root-internally, as a constituent of an unanalyzable primitive. It was added to **de-* as a derivational extension suffix, converting a directional noun into a verbal stem. The standard treatment presents **deh₃-* as an atom. The framework reads it as a derived form: **de-* (hand) + **-h₃-* (verbal extension) = the act of handing over. The laryngeal is explained, not stipulated.

The decomposition has structural symmetry with **-tero-*. In both cases, the bare **de-* is extended by a morpheme (**-r-* or **-h₃-*) that yields a derivative form whose function is predictable from the source-noun’s hand-content: **de-* + **-r-* = the spatial-direction adjective (where the hand points); **de-* + **-h₃-* = the give-verb (the act of handing). **der-* and **deh₃-* are siblings: parallel derivatives of the same bare pre-PIE hand-element, the nominal and verbal branches of a single etymon that the three-paper project has been documenting.

The Mycenaean Greek tablets attest both elements independently: the give-verb *di-do-si* /*didonsi*/ “they give” and the directional clitic *-de* appear in the same corpus. We do not claim that Mycenaean speakers were synchronically aware of the etymological connection — by 1400 BCE the two had been morphologically separate for a very long time. We note only that both elements are independently attested in Greek’s earliest documented form, which is consistent with the framework’s claim that they share a common atomic source.

6. Cross-cultural typology

The framework's typological predictions for this paper are HAND > DIRECTIONAL (underlying the bare **de-* of §2) and HAND > GIVE (underlying **deh₃₋* of §5). Both are well-attested in the cross-linguistic record.

HAND > DIRECTIONAL is a natural specialization of the well-documented HAND > LOCATIVE pathway (Heine & Kuteva 2002). The step from “at the hand of” to “to, toward” is a directional specialization of the locative reading: a spatial adposition with directional rather than stative semantics. Attested cases include P'urhepecha, which forms the locative from the noun *jak'i* “hand” via the suffix *-k'u*, and Mayan languages (Kaqchikel, Tz'utujil, K'iche') where body-part terms function productively as relational nouns expressing spatial locations.

HAND > GIVE is more directly attested. Heine & Kuteva (2002: 167-168) document HAND > AGENT MARKER and related developments; the underlying step HAND > HAND-OVER, GIVE is among the most natural verbal grammaticalizations from a body-part noun. The semantic chain is immediate: to hand something is to give it. Languages where “give” derives transparently from a hand-noun or hand-related expression are documented across language families, making this one of the most typologically robust grammaticalization pathways available to the framework.

7. Future research

The framework has scope beyond the four items (**de-*, **der-*, **tero-*, **deh₃₋*) developed here. We collect the most significant extensions, each requiring its own paper-length treatment.

****dher-* “hold, maintain” and Sanskrit *dharma*.** The verbal root **dher-* is reflected in Sanskrit *dhārayāti* “holds, maintains,” Old Iranian *dāraya-* “holds” (preserved in the royal name Darius: “he who holds firm”), Greek *thrónos* “seat, throne” (if from this root, via “that which holds”), and most consequentially in Sanskrit *dharma* “cosmic order, law, righteousness.” The framework reads **dher-* as a further derivative of **de-*: the hand that holds rather than the hand that gives, yielding the semantic chain HAND > HOLD > MAINTAIN > COSMIC ORDER. The downstream consequence that is hardest to ignore: *dharma*, on the framework's reading, is “that which the hand keeps.” The step from a physical maintenance verb to a principle of cosmic holding-together is documented in the religious vocabularies of multiple language families; the Sanskrit chain from **dher-* to *dharma* is among the most consequential instances of it. A full treatment requires engagement with the Vedic and later Indic literature on *dharma*, the Buddhist and Jain philosophical extensions, and the comparative theology of Indo-European maintenance vocabulary. We defer this to a future paper while noting that it is the framework's most consequential downstream implication: if the framework is correct, the word for cosmic order in the oldest continuous religious tradition in the world has a hand at its etymological root.

PIE **terh₂₋* “to cross over.” This verbal root gives Latin *trāns* “across,” English *through*, Sanskrit *tārati* “passes, overcomes,” Sanskrit *tirás* “across, sideways, beyond,” Hittite *tarḫu-* “prevails.” Sanskrit *tirás* is particularly suggestive: a bare *t-r* form with directional meaning. Whether **terh₂₋* is etymologically related to **der-* through a chain like HAND > REACH-ACROSS > CROSS-OVER is a question we leave open but flag as promising terrain.

The broader hand-semantics network. If the framework is correct, pre-PIE **de-* “hand” is a node of substantial breadth, with reflexes in lexical hand-words (Paper 1), the contrastive suffix (Paper 2), the directional particle and the give-verb (the present paper), and verbal extensions including **dher-/dharma*, **terh₂₋*, and the destructive-action **der-* “flay, split” (Greek *dérō*, Sanskrit *dṛnāti*, Old English *teran*). Mapping the full network is warranted and deferred.

8. Remaining problems

Four problems deserve explicit statement, none of which we regard as fatal.

First, the standard etymology of Greek *-δε* does not unanimously trace it to PIE **de/do*; alternative

reconstructions exist (**eh*₂, **h*₂*e*, **o*) on which -δ*ε* would not be a direct reflex of bare **de*-. The χαμαί case, central to the **eh*₂ reconstruction, is contested (§2.3); we have argued in outline but not exhaustively that the locative reading of χαμαί is preferable. A fuller defense is needed.

Second, Lipp’s (2009) reconstruction of athematic locatives in **er* and **én* is a specific proposal within the heteroclitic-stem literature, not a universally accepted result. If Lipp’s reconstruction is rejected or modified, the *-*r*- of the framework loses its primary independent attestation as a locative morpheme. The framework would still be consistent with alternative analyses, but the specific compositional analysis of §4 would need adjustment.

Third, the property suffix *-*ro*- (in **rudh*-ró- “red”) and the instrument suffix *-*tro*-/*trom* (in **méd*-*trom* “measure”) contain *-*r*- but do not have obvious locative semantics. If these are the same morpheme as Lipp’s locative **er*, the framework needs an account of the semantic divergence. If they are homophonous separate morphemes, the framework needs to motivate the homophony. We leave this open.

Fourth, the proposed function of the laryngeal extension *-*h*₃- in **deh*₃- (§5) is reconstructed rather than directly motivated. We have appealed to a general stative/factitive function of *-*h*₃- in PIE morphology, but the specific semantic contribution of this extension to the verbal stem deserves fuller engagement with the laryngeal literature.

A fifth problem deserves explicit statement because it concerns the epistemological status of the paper’s central claim rather than any specific empirical difficulty. The identification of PIE **de*/*do* “to, toward” as a reflex of the pre-PIE hand-noun rests primarily on typological plausibility — the HAND > DIRECTIONAL pathway is cross-linguistically attested, and the semantic profile fits the framework’s predictions. Typological plausibility is not uniquely identifying evidence; many languages have directional particles with no connection to hand-nouns. The identification would be weakened — possibly decisively — by: (a) a convincing alternative etymology for PIE **de*/*do* that accounts for the attested reflexes with equal or greater parsimony; (b) evidence that Melchert’s (2017) reconstruction of the Greek-Anatolian allative from PIE **eh*₂ is exhaustive, leaving no role for a **de*-derived allative element; or (c) cross-linguistic evidence that the HAND > DIRECTIONAL pathway is blocked in the environments where **de*/*do* appears. We do not know of evidence in any of these categories at present. The identification is falsifiable in principle, and the conditions just listed are where the falsifying evidence would have to come.

9. Conclusion

Across three papers this project has argued for a single substantive proposal: that pre-PIE had a body-part etymon meaning “hand, open palm,” and that this etymon underlies a network of lexical, morphological, and verbal reflexes in PIE whose unity the standard handbooks have not recognized. Paper 1 documented the lexical branch: the cognate cluster Greek δῶρον/Hesychian δῶρις/Celtic **durno*-/Latvian **dũre* reflects a body-part root **der*- “hand.” Paper 2 extended the case into morphology: the contrastive suffix *-*tero*- descends from the same etymon via HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE, and its bilateral character — the property every standard reference documents but none etymologizes — is a structural consequence of what **de*-*r*-*o*-composes to mean.

The present paper has gone one step deeper. **der*- itself is bimorphemic: bare **de*- (the atomic hand-element, independently attested as the PIE directional particle **de*/*do*, most directly in the Mycenaean allative clitic -*de* productive by 1400 BCE) plus *-*r*- (the locative/spatial morpheme, independently attested in Lipp’s (2009) reconstruction of PIE heteroclitic locatives). The contrastive *-*tero*- decomposes compositionally as **de*- + *-*r*- + *-*o*-, with each piece independently attested and the resulting semantics emerging transparently from the parts. The same bare **de*- underlies the give-verb **deh*₃- = **de*- + *-*h*₃-: the nominal branch (**der*-) and the verbal branch (**deh*₃-) are siblings, not competitors.

The framework makes a specific etymological commitment at each step where the standard view

leaves the question open. **de/do* is either a primitive PIE particle of unknown origin (standard) or the bare descendant of a pre-PIE hand-noun (framework). **-r-* is either a heteroclitic alternant or an extension (standard, multiple incompatible analyses) or a single locative/spatial morpheme (framework, following Lipp). **-tero-* is either a primitive suffix of unknown etymological origin (standard) or **de-* + **-r-* + **-o-* (framework). **deh₃-* is either a primary verbal root (standard) or **de-* + **-h₃-* (framework). At each step the framework provides a unified motivation where the standard view leaves a gap. Whether the commitments are correct is for the field to evaluate. We claim only that they are internally consistent, supported by independent typological evidence, and worth examination.

10. Probability estimate

Following the practice established in Papers 1 and 2, we close with an explicit probability assessment framed conditionally.

Conditional on both Papers 1 and 2 being correct — i.e., that pre-PIE had a body-part word **der-* “hand, open palm” (Paper 1) and that **-tero-* descends from the same etymon via the grammaticalization pathway HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE (Paper 2) — Claude’s estimate of the probability that the substantive thesis of this paper is correct in its basic outline is approximately 30–40%. The conditional estimate is higher than in earlier drafts because the compositional decomposition is a more constraining claim than the convergence argument alone: three independent matches of phonology and semantics (**de-*, **-r-*, **de-r-o-*) converging on the predicted composition is harder to attribute to coincidence than the softer claim that the suffix and the hand-noun share an etymological source.

The marginal probability — unconditioned, integrating over the probability that Papers 1 and 2 are also correct — is approximately the product: 15–20% (Paper 1) × 35–45% (Paper 2 conditional on Paper 1) × 30–40% (Paper 3 conditional on Papers 1 and 2) ≈ 1.6–3.6%. This marginal figure represents the cumulative ambition of the framework accurately and is substantially lower than any individual conditional estimate. We continue to regard the central claim as more likely wrong than right, and also to regard it as interesting enough, and the evidential structure dense enough, that publication for examination by other scholars is warranted.

11. Authorship and contributions

This is the third paper in the *der-* project. It builds on Papers 1 and 2 (Chavis & Claude 2026a, 2026b) and continues the same co-authorship arrangement.

Chavis’s contributions. The substantive intellectual direction of Paper 3 originates with Chavis. The hypothesis that the directional particle **de/do* (specifically as preserved in Greek *-δε*) might descend from a free-standing pre-PIE hand-element is Chavis’s insight. The decision to pursue the bimorphemic decomposition of **der-* itself — separating bare **de-* from the **-r-* extension — is Chavis’s, as is the framing of the paper around compositional evidence. The sharpest question of the paper, “whence comes the *r* in **-tero-*?” — which led to the engagement with Lipp’s reconstruction of athematic locatives and to the compositional analysis of §4 — is Chavis’s. Chavis also identified the contestability of Melchert’s *χαμᾱί*-as-allative reading, and directed the scope reduction of the paper to its four core items (**de-*, **der-*, **-tero-*, **deh₃-*). The comprehensive rewrite that produced v3 was directed by Chavis in response to accumulated structural and prose concerns across the project’s arc review.

Claude’s contributions. Claude’s contributions are primarily in source synthesis and exposition. Claude identified and surfaced the relevant published literature: Lipp (2009) on the heteroclitic locatives; Lazzeroni (2006) and Melchert (2017) on the Greek-Anatolian allative parallel; Beekes (2010) and Frisk (1960) on *χαμᾱί*; Lejeune (1961) and van Hes (2021) on Mycenaean *-de*; Dunkel (2014) on PIE **de/do*; Hackstein (2018) on the allative formations. Claude drafted the prose from Chavis’s structural and substantive direction, proposed the Anatolian distribution argument (§8 of

Paper 2), and is responsible for the specific framings of the arguments and the local rhetorical choices. Claude's probability estimate in §10 is its own.

Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management. The collaboration is genuine. The argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

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